Dhaneswara Mandrasa

Senior Data Scientist & AI Engineer · Portfolio: dhaneswara-mandrasa.vercel.app

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PROFESSIONAL SUMMARY

Senior Data Scientist & AI Engineer with 8+ years across telecom, banking, commerce and geospatial: ML, causal inference and LLMs for pricing, risk and mobility. Three years in Abu Dhabi at Eureka AI working with 7 telecom operators in 7 countries: segmented 100M+ daily users for clients including Gojek, Grab and Tokopedia, built a credit score with the company's best-ever Gini (0.6) and grew the products I led from 5% to 70% of revenue. Shipped a production LLM system on Snowflake Cortex and led data & AI for a 30K-merchant commerce platform. Now building LLM feature-extraction pipelines for causal and demand models at Hiroshima University.

RESEARCH

Master's researcher in Smart Mobility, Hiroshima University. I build LLM feature-extraction pipelines that turn unstructured geospatial text into reliable, validated features for causal and demand models. Current work: using NYC's January 2025 congestion toll as a natural experiment on millions of ride-hail trips, I test whether LLM-scored place features improve price-response and welfare estimates against a no-LLM baseline.

LLM features for causal pricing evaluation · Urban and Data Science Lab, Hiroshima University **2026 – present**

- **LLM feature extraction / context engineering:** converted OpenStreetMap POIs and place names into structured prompts, and scored 15 semantic features across 4 construct families for 261 NYC zones with per-trait prompts and three LLMs (Llama-3.3 70B, Qwen-2.5 72B, GPT-4o-mini).
- **LLM evaluation and reliability harness:** built a versioned, hashed prompt library and a generalizability study that separates zone signal from wording and model noise (20,250 pilot scores, 0 invalid); 14 of 15 features reach G ≥ 0.85, with pre-2025 training cutoffs ruling out leakage.
- **Causal impact / geo-experiment:** ran a difference-in-differences (incrementality) analysis of the congestion toll on millions of NYC TLC high-volume trip records and estimated a −2.6 to −2.9 pp shift in ride-hail share in tolled zones.
- **Ablation:** do LLM features add value? Designed an ablation ladder (no-LLM baseline → LLM features → richer representations) that measures incremental lift; estimation in progress.
- **Demand and price elasticity modelling:** building a share-logit demand model whose price coefficient gives price elasticity and willingness-to-pay by zone segment (in progress).
- **Data engineering:** built a bronze/silver/gold pipeline over millions of trip records.

PROFESSIONAL EXPERIENCE

Olsera – cloud POS & commerce platform, 30K+ merchants across Indonesia **Jakarta, Indonesia**
Head of Data & AI Aug. 2025 – Mar. 2026

- Owned the data & AI roadmap for a platform processing hundreds of thousands of transactions a day, reporting to executives; hired and led the data team.
- Built the data function from scratch: architecture and schemas for orders, products, roles and logs behind analytics and AI use cases.
- Led the platform decision with proof-of-concept evaluations of Snowflake vs Databricks, benchmarked against BigQuery and Redshift; prototyped AI stock recommendations for merchants.

iZeno – enterprise IT & data consulting **Jakarta, Indonesia**
Lead Data Scientist Jul. 2024 – Mar. 2025

- Led and delivered enterprise AI/ML projects for clients across financial services, agriculture and banking as part of iZeno's data consulting practice.
- Cut the run time of a credit-scoring data and ML pipeline by 75%.
- Built a production LLM news-analytics system on Snowflake Cortex for a financial-sector client, cutting misclassification by 27% with a knowledge graph of 10K+ entities.
- Designed a hybrid recommendation engine for a bank that improved targeting accuracy by 35%, and worked with domain experts on a root-cause analysis system (+18% yield).

Eureka AI – telecom data & AI, 7 operators in 7 countries **Abu Dhabi, UAE**
Senior Data Scientist Apr. 2021 – Jul. 2024

- Scale: worked with 7 telecom operators in 7 countries (Indonesia, Malaysia, Singapore, Thailand, Vietnam, Hong Kong, UAE) on hundreds of millions of rows, in a multinational team from Singapore, Indonesia, Malaysia, India, the UAE and the US.
- Segmentation: segmented 100M+ daily active ride-hailing and e-commerce users into hierarchical segments with supervised and unsupervised models (k-means, Random Forest, XGBoost) on clickstream data in PySpark ML and MLflow; models served 5 major digital-platform clients including Gojek, Grab, Tokopedia and foodpanda.

- Personas: identified 30+ personas for 80M+ monthly active telecom subscribers from clickstream, mobility and subscriber data, with a Word2Vec look-alike model in Spark; both became reusable products across clients.
- Credit risk: built an alternative credit score from telco and clickstream data reaching a Gini of 0.6, the highest in the company's history.
- Product & revenue: grew the Market Intelligence products from 5% to 70% of company revenue in under two years; presented models to the Chief Product and Chief Commercial Officers and supported the commercial team as technical lead in client pitches.
- Team: hired, mentored and set modelling standards for a data science team of 6; made models explainable to business users with SHAP.

Bhumi Varta Technology – location intelligence **Jakarta, Indonesia**
Data Scientist Jun. 2020 – Apr. 2021

- Built spatial models on POI, mobile and sales data that guided clients' expansion from under 100 to 200+ outlets across Indonesia.
- Estimated poverty levels from satellite imagery with CNNs (PyTorch), refining socio-economic status resolution from 1×1 km to 100×100 m.

AEON Indonesia – retail **Jakarta, Indonesia**
Management Trainee Apr. 2018 – Sep. 2019

- Ran sales and operations analysis during a cross-functional rotation (store operations, merchandising, marketing, supply chain), turning store data into process-improvement recommendations.

SKK Migas – upstream oil & gas regulator **Jakarta, Indonesia**
Data Analyst Apr. 2016 – Apr. 2018

- Built an early ML model predicting oil & gas resources for 200+ working areas (<10% test error); migrated data from 50+ companies; built Python/Tableau dashboards.

EDUCATION

Hiroshima University **Hiroshima, Japan**
MPhil, Smart Mobility (AI and data science for transport) Apr. 2026 – Mar. 2028 (expected)
Urban and Data Science Lab, supervised by Prof. Tao Feng. Programme: Innovation and Practice for Smart Society.

Institut Teknologi Bandung (ITB) **Bandung, Indonesia**
B.Sc. Geological Engineering – GPA 3.52/4.00, Cum Laude Aug. 2011 – Aug. 2015

TECHNICAL SKILLS

Programming: Python (pandas, polars, NumPy, scikit-learn, statsmodels), SQL, R, PySpark

AI & LLM stack: LangChain, LangGraph, Snowflake Cortex, Llama-3.3, Qwen-2.5, RAG (Elasticsearch), prompt design and evaluation, knowledge graphs, FastAPI, spaCy, PyTorch

Machine learning: supervised and unsupervised learning, XGBoost and gradient boosting, Random Forest, clustering and segmentation, recommendation systems, credit scoring, CNNs, SHAP, MLflow

Data platforms: Snowflake, Databricks, Apache Spark / Spark ML, Dataiku, Airflow, medallion (bronze/silver/gold) pipelines

Cloud & DevOps: AWS, GCP, Azure, Docker, Git

Causal inference & econometrics: difference-in-differences, discrete choice (MNL, NL, ICLV) in Biogeme, factor analysis, price elasticity and welfare estimation

Geospatial: QGIS, GeoPandas, OpenStreetMap, satellite imagery, POI and mobility data

LANGUAGES

English (advanced); Bahasa Indonesia (native)